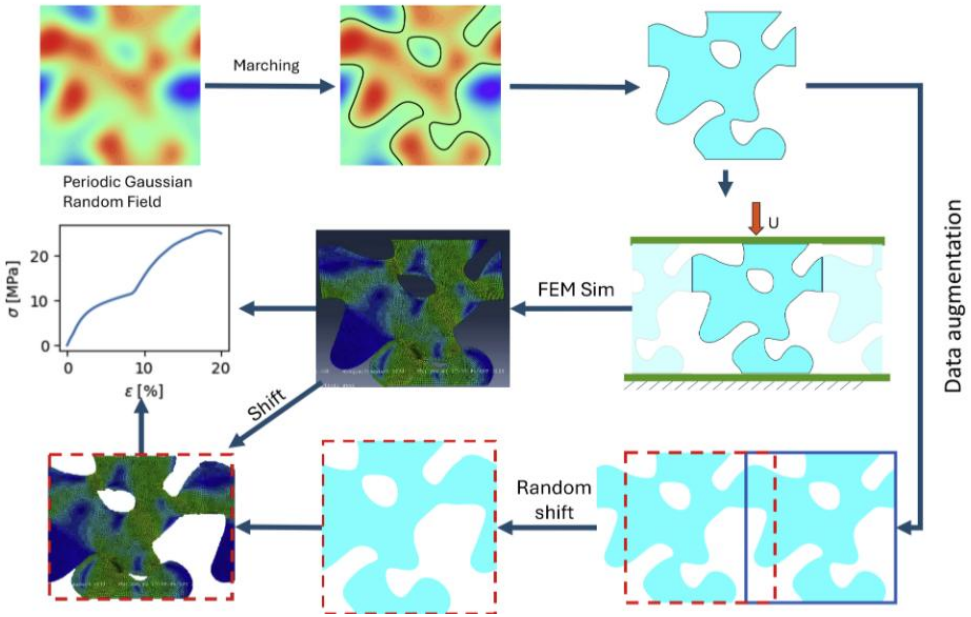
Statement of Problem

- The inverse design of metamaterial unit cells aims to **generate geometries that achieve a target stress–strain response**.



VAR – Implementation Details

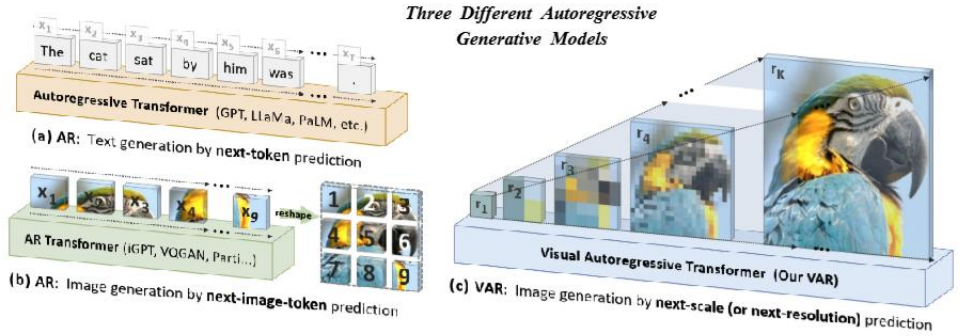


Figure 2: **Standard autoregressive modeling (AR) *vs.* our proposed visual autoregressive modeling (VAR).** (a) AR applied to language: sequential text token generation from left to right, word by word; (b) AR applied to images: sequential visual token generation in a raster-scan order, from left to right, top to bottom; (c) VAR for images: multi-scale token maps are autoregressively generated from coarse to fine scales (lower to higher resolutions), with parallel token generation within each scale. VAR requires a multi-scale VQVAE to work.

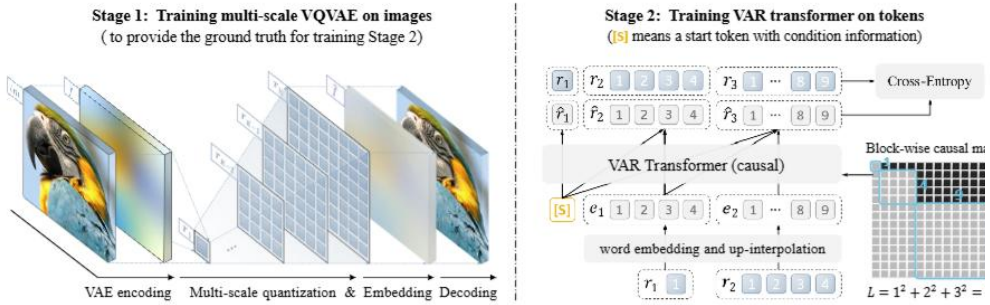


Figure 4: **VAR involves two separated training stages. Stage 1:** a multi-scale VQ autoencoder encod an image into K token maps $R = (r_1, r_2, \ldots, r_K)$ and is trained by a compound loss (5). For details "Multi-scale quantization" and "Embedding", check Algorithm 1 and 2. **Stage 2:** a VAR transformer is train via next-scale prediction (6): it takes $(\texttt{[s]}, r_1, r_2, \ldots, r_{K-1})$ as input to predict $(r_1, r_2, r_3, \ldots, r_K)$. Th attention mask is used in training to ensure each r_k can only attend to $r_{\leq k}$. Standard cross-entropy loss is used

$$p(x_1, x_2, \ldots, x_T) = \prod_{t=1}^{T} p(x_t \mid x_1, x_2, \ldots, x_{t-1}). \tag{1}$$

$$f = \mathcal{E}(im), \qquad q = \mathcal{Q}(f), \tag{2}$$

$$q^{(i,j)} = \left(\underset{v \in [V]}{\arg\min} \|\text{lookup}(Z, v) - f^{(i,j)}\|_2 \right) \in [V], \tag{3}$$

$$p(r_1, r_2, \ldots, r_K) = \prod_{k=1}^{K} p(r_k \mid r_1, r_2, \ldots, r_{k-1}), \tag{6}$$

$$L = \log p(x|z_q(x)) + \|\text{sg}[z_e(x)] - e\|_2^2 + \beta \|z_e(x) - \text{sg}[e]\|_2^2,$$

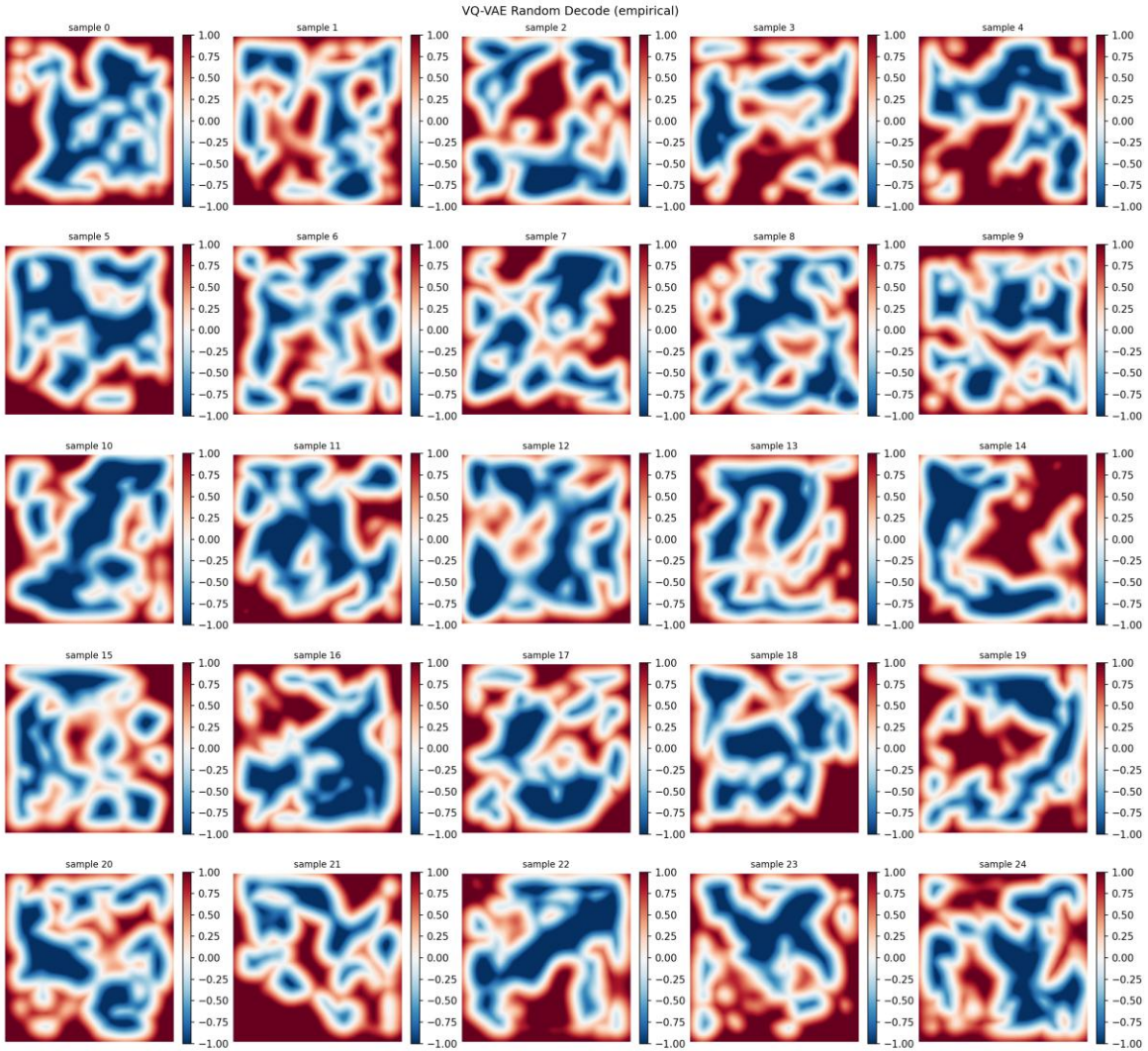
Algorithm 1: Multi-scale VQVAE Encoding

Inputs: raw image im;
Hyperparameters: steps K, resolutions $(h_k, w_k)_{k=1}^{K}$;
$f = \mathcal{E}(im)$, $R = []$;
for $k = 1, \cdots, K$ **do**
  $r_k = \mathcal{Q}(\text{interpolate}(f, h_k, w_k))$;
  $R = \text{queue_push}(R, r_k)$;
  $z_k = \text{lookup}(Z, r_k)$;
  $z_k = \text{interpolate}(z_k, h_K, w_K)$;
  $f = f - \phi_k(z_k)$;
Return: multi-scale tokens R;

Algorithm 2: Multi-scale VQVAE Reconstruction

Inputs: multi-scale token maps R;
Hyperparameters: steps K, resolutions $(h_k, w_k)_{k=1}^{K}$;
$\hat{f} = 0$;
for $k = 1, \cdots, K$ **do**
  $r_k = \text{queue_pop}(R)$;
  $z_k = \text{lookup}(Z, r_k)$;
  $z_k = \text{interpolate}(z_k, h_K, w_K)$;
  $\hat{f} = \hat{f} + \phi_k(z_k)$;
$\hat{im} = \mathcal{D}(\hat{f})$;
Return: reconstructed image $\hat{im}$;

VAR Results and Learning Reflections



- Learned inverse neural operators and most recent generative models including autoregressive advancements and diffusion